Plane mirrors

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use the reflection law and locate a virtual image.

SUCCESS CRITERIA - I can...

1. Measure angles from the normal.
2. Use equal incidence and reflection angles.
3. Place the image the same perpendicular distance behind the mirror.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “normal line” and “reflection” to explain your model or result. Sentence starters are optional.

normal line: a reference line perpendicular to a surface

reflection: a wave returning from a boundary

virtual image: an image at an apparent origin of rays, not their meeting point

01 Warm-up: bring back an earlier idea

Sensors 1.02 m apart record delay 0.003 s.
Find wave speed. If 0.0005 s is added to the delay, find the incorrect speed and describe the bias.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

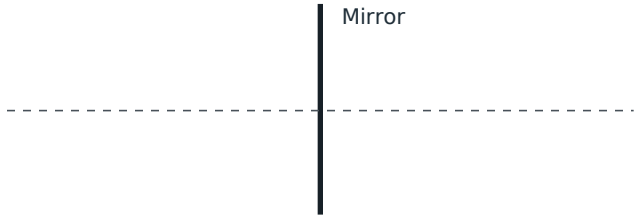
02 Worked example: explain the model

Reference: $\theta_{incident} = \theta_{reflected}$
image distance behind = object distance in front.
Plane mirror; measure angles from the normal. The image is virtual.

Worked reasoning: An object 2 m in front of an ideal plane mirror has a virtual image 2 m behind it. The object-image separation is 4 m; reflected rays appear to come from the image.

Your turn: Draw the mirror, object, image and backward extensions of two reflected rays.

Sentence starter: The … in my model represents … . Its label shows … .



Add object / image or incident / reflected rays.

Marzano check - circle: ① ② ③

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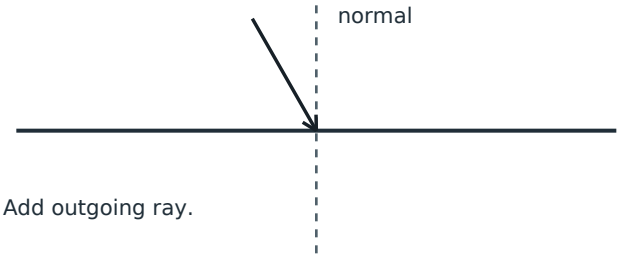
Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw the normal at the mirror. Complete a reflected ray for an incident ray 30° from normal.

Sentence starter: The ... in my model represents ... . Its label shows ... .



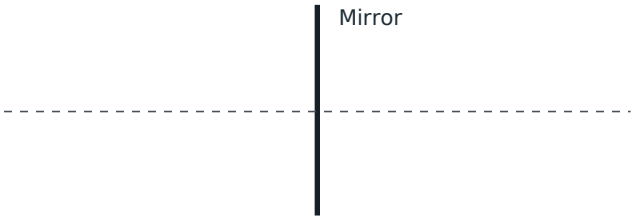
Add outgoing ray.

Marzano check - circle: 1 2 3

04 Apply the physics

A laser ray makes a 20° angle with a plane mirror's surface.
Draw the normal and both rays.
Find incidence angle measured from the normal, then the angle between the incident and reflected ray segments drawn away from the contact point.

Sentence starter: My design shows ... . It meets the requirement by ... .



Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why is the mirror image called virtual rather than a real object behind the glass?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

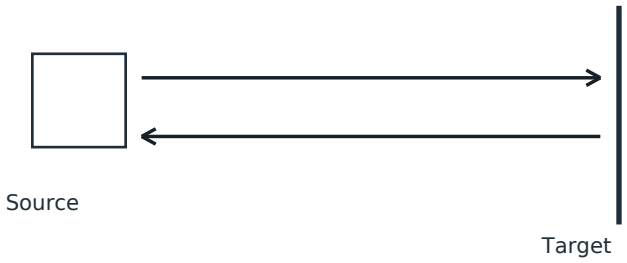
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

Echoes return at 0.004 s and 0.006 s with the same wave speed.
Compare ranges. Does delay alone determine mass?
Label the outgoing and returning paths.

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3

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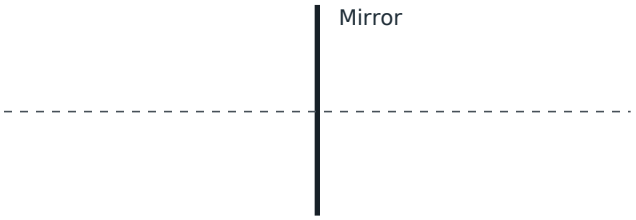
Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

An object moves 0.50 m toward a fixed mirror. **Find** image movement and change in object-image separation.

Sentence starter: I found ... [unit]. This value means ... in this situation.



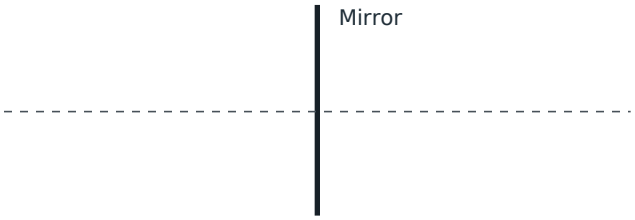
Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

An object moves toward a fixed plane mirror at 1.2 m/s. **Find** image velocity relative to the mirror and the rate at which object-image separation decreases.

Sentence starter: My design shows ... . It meets the requirement by ... .



Add object / image or incident / reflected rays.

Marzano check - circle: 1 2 3